Decision-Task Portability of PV Characterization Sequences: A Retrospective Cross-Climate Audit

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Abstract

Whether an existing photovoltaic (PV) characterization sequence can be reused across climates depends on the decision task. We retrospectively reweight a fixed nine-product, 27-condition laboratory response panel with operating-state distributions measured on different PV systems; these are not new field tests of the nine products. Using unchanged pair-level stopping records, we compare mean pairwise depth, accepted-winner identification, and shared all-pairs campaign burden. For Cocoa-to-Eugene transfer, mean depth rises from 8.08 to 8.75 cells and the all-pairs prefix from 14 to 17, while the correct accepted winner is identified at depth 8 instead of 14. Among 190 historical-order replays on 38 external targets, four accepted winners disagree with the complete reference. Twelve of 184 correct-winner, all-pairs-complete replays show strict task reversals; nine belong to the Brazilian source family and nine change winner depth by one cell. Of 79 mean-improving runs, 21 worsen both campaign proxies. Graph bounds explain why mean-depth improvement cannot guarantee campaign savings. A separate ten-unit fixed-calibration-bank sensitivity shows no task reversal and exposes two residual-envelope breaches. Nominal source-shortlist exclusions are a secondary set-valued diagnostic. These finite retrospective results are not prospective error guarantees or measured laboratory-time savings.

Keywords: campaign burden; decision-task portability; photovoltaic module characterization; shortlist transfer; winner identification

1 Introduction

Even when a photovoltaic (PV) module remains preferred in two climates, the portability of the evidence used to support that preference depends on what the evidence must accomplish. A fixed characterization sequence may be evaluated as a mean over all pairwise decisions, as a procedure for identifying only the best product, or as a shared campaign that must finish every comparison. These objectives are not equivalent. A separate source shortlist can also remove target-relevant candidates. This retrospective study therefore asks a task-conditioned question: when an existing laboratory evidence path is reweighted for another location's operating states, does its burden or screening consequence change for the declared decision task, and does the reverse direction behave differently?

Multi-site field studies describe climate-dependent performance for matched PV technologies or configurations [1–4]; the International Energy Agency Photovoltaic Power Systems Programme (IEA PVPS) Task 13 addresses climate-specific selection, monitoring and data

quality [5, 6]. PV parameter extraction from manufacturer datasheets [7] addresses a different kind of sparse evidence. International Electrotechnical Commission (IEC) standards 61724/61853 and recent studies cover monitoring, characterization and measurement uncertainty [8–17]. A Gulf Cooperation Council (GCC) within-matrix decision framework [18] likewise differs from between-climate evidence transport. Research on ranking-and-selection and safe screening distinguishes information burden from false elimination [19–24]; the cited studies do not directly evaluate portability of these two PV evidence-reuse operations.

The measured quantity underlying the laboratory branch is source-tabulated, corrected maximum-power-point power, P_{mp}, at specified irradiance ($\mathrm{W\,m^{-2}}$) and module temperature (°C). STC normalization produces dimensionless response values; interpolation of independently observed G_{POA}–T_{mod} states supplies the exposure weights, not new measurements of those products. A reference-pair residual envelope determines the retrospective stopping event. None of these steps supplies traceable cell-level uncertainty without further instrument calibration records.

The closest comparisons help define the scope. Blakesley et al. [13] analyze the accuracy and cost of reduced IEC 61853 energy-rating measurements with an explicit laboratory uncertainty model. That is distinct from asking whether an *already specified characterization sequence* retains its decision burden when moved from a source to a target climate. Marion et al. [1] compare observed PV-unit yields across sites, but do not report the complete directional shortlist-exclusion intervals evaluated here. The present contribution is a bounded *audit of evidence reuse*, not a new measurement optimizer, a measured laboratory-time saving, or a metrologically calibrated screening guarantee.

The Real Testing Conditions (RTC) framework derives site-representative irradiance, temperature, and spectral conditions for PV benchmarking [25]. Unlike RTC, our analysis holds the response grid fixed and asks whether the *ordered evidence path* and a source shortlist retain their decision consequences after transfer; it does not replace IEC qualification.

The study examines these two questions with separate analyses because they reuse different kinds of evidence. In *Stage A*, fixed test orders for nine public PV response matrices are replayed under observed operating-state weights with an empirical stopping rule. The endpoint is a count of model-evaluated test cells, not laboratory minutes or calibrated wrong-decision risk. In *Stage B*, published field yields are used to check whether a source-site shortlist excludes modules still competitive at the target. The primary stages use distinct product panels, not consecutive steps of one experimentally tested workflow. A separate ten-unit subcohort uses a fixed calibration bank of different Golden physical modules to test laboratory sequence reuse without equating noncontemporaneous laboratory and field endpoints. External operating-state records extend the Stage-A target conditions but do not measure the nine reference products at new sites; support gaps, wrong accepted signs, unresolved cases and dependent windows remain explicit.

The directed results show that "portable" is not a single property of a sequence. In the original Cocoa→Eugene transfer, the source order is worse than the target-conditioned heuristic for mean pairwise depth but better for identifying the accepted winner. Across the external targets, finite objective reversals also occur, while the restricted Golden-bank sensitivity shows that such reversals are not universal across cohorts. Stage B remains a separate screening operation: a 5% Cocoa-derived shortlist removes three of five Eugene-relevant exact units. The contributions are therefore three bounded, task-specific results:

1. *Decision-task separation:* the same pairwise stopping records are evaluated for mean pair resolution, accepted-winner identification, and shared all-pairs campaign completion; their portability conclusions can disagree.

2. *Campaign-burden structure:* simple bounds and a finite counterexample show why a

lower pairwise mean cannot by itself guarantee a shorter shared campaign; 21 of 79 mean-improving external runs worsen both campaign proxies.

3. *Secondary shortlist extension:* exact-unit and matched-configuration field panels disclose directional threshold intervals in which source screening excludes target-relevant candidates, without pooling the product cohorts.

2 Materials and Methods: Decision-Task Evidence Transport

The two branches test whether an evidence-reuse choice made for a source climate remains suitable at a target climate. They differ in what is reused: Stage A transports the *order of characterization conditions*, whereas Stage B transports the *set of candidates retained for comparison*. In the primary audits the products and measured endpoints are different, and no combined prospective within-product transfer procedure is evaluated; a smaller retrospective subcohort with shared unit identities is reported separately below.

2.1 Stage A: characterization-order transport

Stage A evaluates a product's *fixed* response profile under a target climate's operating states. The analysis uses the public Sandia PVPMC IEC 61853-1 matrix dataset for nine modules (IDs 19074-001 through 19074-009), reduced to 27 retained irradiance–temperature nodes per product [26]. For response-panel product p and environment c, define the dimensionless weighted score

$$S_{p,c} = \sum_{j=1}^{27} w_{j,c} z_{pj}, \qquad w_{j,c} \geq 0, \quad \sum_j w_{j,c} = 1, \tag{1}$$

where j indexes grid nodes, z_{pj} is node power divided by product p's measured power at standard test conditions (STC), and $w_{j,c}$ is the mean barycentric weight from admitted plane-of-array irradiance (G_{POA}) and module-temperature (T_{mod}) states. The score in Eq. (1) is dimensionless, and field records provide weights, *not* new performance labels for the nine products.

To define which of two products is preferred under those target weights, let a and b index a held-out product pair. Their full-response contrast is

$$\Delta_{ab,c} = \sum_j w_{j,c}(z_{aj} - z_{bj}). \tag{2}$$

Under Eq. (2), positive $\Delta_{ab,c}$ favors a, negative favors b, and zero is a full-reference tie. The fixed source order π_s ranks non-STC nodes with weight $> 10^{-14}$ by descending source weight, breaking ties by irradiance and temperature (Supplementary Section 13). After k non-STC nodes, $\widetilde{\Delta}_{ab,c}(k; s)$ uses only visited responses. The other seven fully characterized products supply 21 calibration-pair residuals for the nonnegative, dimensionless envelope $B_{k,c}^{(a,b)}$; the held pair's unseen full contrast is excluded. Stop at the first k satisfying

$$\left|\widetilde{\Delta}_{ab,c}(k; s)\right| > B_{k,c}^{(a,b)}. \tag{3}$$

Crossing Eq. (3) accepts an empirical sign; exhaustion without crossing is unresolved. The envelope is not a metrological confidence interval. STC is mandatory and included in

depth. For 36 fully resolved pairs, let $\bar{k}_{s\to c}$ be mean source-order depth and $\bar{k}_{c\to c} > 0$ the target-conditioned heuristic mean. Percentage excess is

$$E^{\mathrm{acq}}_{s\to c} = 100\left(\frac{\bar{k}_{s\to c}}{\bar{k}_{c\to c}} - 1\right). \tag{4}$$

In Eq. (4), zero excess means equal mean stopping depth; positive values require more test cells than the target-conditioned reference, while negative values require fewer. The denominator is the *target* reference, so the two transfer directions need not agree. Neither negative excess nor strict-stop resolution establishes optimality or calibrated error risk.

2.2 Decision tasks and task-specific burden

Let m denote the number of compared products and k_{ab} the accepted stopping depth for resolved pair (a, b) under one fixed sequence. The original Stage-A endpoint is the edge average

$$\bar{k} = \binom{m}{2}^{-1} \sum_{a<b} k_{ab}. \tag{5}$$

For a shared campaign that completes all pairwise decisions, define

$$K_{\mathrm{all}} = \max_{a<b} k_{ab}, \qquad N_{\mathrm{mod-cell}} = \sum_{a=1}^{m} \max_{b\neq a} k_{ab}. \tag{6}$$

The first quantity is the common prefix required before every pair is resolved; the second is an idealized early-release count in which each product is measured only until its latest required incident comparison. These are planning proxies, not elapsed laboratory time.

A different task is to identify only the best product. At each accepted pairwise stop, the empirical sign defines a directed win. A product is an *accepted winner* when it has accepted wins against all other $m-1$ products. Its accepted-winner depth K_{win} is the largest stopping depth among those incident wins; no accepted winner is reported if such a unique product never appears. Correctness is labeled only afterward against the full weighted-score winner. Accordingly, a smaller K_{win} is an accepted-decision *speed* observation, not a valid benefit unless the accepted winner is retrospectively correct. That correctness gate is unavailable to a prospective user without full-reference evidence and is not a safety certificate. This is a stricter identification requirement than eliminating a product after one loss and uses no unseen held-pair response in the stopping rule.

These task metrics are structurally different. Writing $r_a = \max_{b\neq a} k_{ab}$ gives the deterministic bounds

$$\bar{k} \le K_{\mathrm{all}}, \qquad m\bar{k} \le N_{\mathrm{mod-cell}} \le mK_{\mathrm{all}}. \tag{7}$$

The lower bound follows from $k_{ab} \le (r_a + r_b)/2$ and summation over the complete comparison graph. Componentwise depth dominance, $k^{(A)}_{ab} \le k^{(B)}_{ab}$ for every pair, makes all three all-pairs metrics no worse. A lower *mean* alone does not: with three products, depths $(1, 1, 10)$ have mean 4, $K_{\mathrm{all}} = 10$ and $N_{\mathrm{mod-cell}} = 21$, whereas $(5, 5, 5)$ have mean 5, $K_{\mathrm{all}} = 5$ and $N_{\mathrm{mod-cell}} = 15$. Likewise, there is no monotone implication from $\bar{k}$ to K_{win}. For $m = 2$, however, there is only one pair, so $\bar{k} = K_{\mathrm{all}} = K_{\mathrm{win}} = k_{12}$ whenever an accepted winner exists, and $N_{\mathrm{mod-cell}} = 2k_{12}$. Thus at least three products are required for strict pair-mean versus winner-depth disagreement. Sequence portability is evaluated against an explicitly declared task rather than a single scalar burden.

2.3 Measurement workflow and audit states

Stage A has a pair-specific *information contract*. The field G_{POA}–T_{mod} records determine supported target weights, not new power measurements of the nine response-panel products. For each evaluated pair, its seven other products supply fully characterized 27-node calibration profiles. The held pair reveals only the STC-normalized conditions already visited in the fixed sequence; the strict stopping rule never uses that pair's unseen full-response contrast. The seven full calibrator profiles are pre-existing evidence; rotating all 36 held pairs presupposes the nine-product reference archive. Acquiring that archive from scratch has unmeasured cost and cannot be counted as saved laboratory effort (Supplement). An independent sequential shadow replay reproduced all *5,400* archived target-local pair decisions across 150 supported target distributions using this information restriction (Supplement). A wrong or correct accepted sign is assigned only afterward against the full target reference; an unresolved pair is detectable once eligible conditions are exhausted. The endpoint counts test cells, *not* laboratory minutes, cost or calibrated uncertainty. Mean excess is comparable only if all 36 pairs resolve. The audit retains wrong signs, grid support and dependent windows; it is *not* a prospective safety guarantee.

Computational assistance and verification. OpenAI ChatGPT assisted in drafting and reviewing analysis, verification, and figure-generation scripts. Ahmed Elsayed personally reviewed the analysis and verification code he worked on and checked numerical outputs against reported results and verification records. The authors retain full responsibility for the work. The same-code raw-input replay and the separate output re-aggregation are documented in the Supplementary Material; neither constitutes independent physical validation.

2.4 Disjoint Golden-bank sensitivity and information contract

The restricted same-unit study uses ten Cocoa–Eugene modules with source-measured, corrected P_{mp} values on an identical 18-condition grid; `aSiMicro03036` lacks this grid and remains in the original 11-unit field-yield analysis only. The 18 points are a source-specific subset, not the full normative IEC 61853-1 measurement matrix. Ten *different physical* Golden modules provide a fixed, fully measured calibration library. They belong to related manufacturer/model families in the same NREL producer archive; disjoint serial identity does not imply independent instrument errors. In the fixed-bank test, all 45 target pairs among the ten Cocoa–Eugene modules use the *same* 45 Golden calibration-pair residuals. The original rotating-bank comparator instead uses the eight non-held target modules for each pair and is reported separately.

Let $\mathcal{G}$ denote the ten Golden calibrators, $z_{u,j} = P_{u,j}/P_{u,\mathrm{STC}}$ their dimensionless responses, and $\mathcal{V}_k(s)$ the non-STC nodes visited by source order s after k steps. For a target climate c, the fully measured calibration contrast minus its visited-node partial contrast gives the empirical envelope

$$r^{\mathcal{G}}_{uv,k,c}(s) = \sum_{j \notin \mathcal{V}_k(s)} w_{j,c}(z_{u,j} - z_{v,j}). \tag{8}$$

$$B^{\mathcal{G}}_{k,c}(s) = \max_{u<v \in \mathcal{G}} \left| r^{\mathcal{G}}_{uv,k,c}(s) \right|. \tag{9}$$

For each held test pair, the strict rule in Eq. (3) uses this bank maximum in place of its rotating-pair envelope; held-pair unvisited responses enter only the subsequent full-laboratory-reference sign audit. STC is acquired and counted in the reported depth; its normalized pair contrast is zero. The Golden profiles comprise $10 \times 18 = 180$ existing module–condition responses whose acquisition and maintenance costs are *not* included in

the stopping-depth proxy. Site weights average five matched G_{POA}–T_{mod} histories per site, after restricting each to the 18-node convex hull. The 56 banks of size 8–10 are overlapping subsets of these same ten Golden profiles, not independent validations (Supplementary Tables S8–S10). A calibration maximum is not assumed to bound a previously unseen test pair.

A retrospective comparator holds these inputs fixed while evaluating 5,000 seeded non-STC order permutations per target and the 136 single-swap neighbors of each local weight-descending order. The swaps use complete held-pair outcomes and are not prospective policies (Supplementary Table S11).

2.5 Stage B: shortlist transport

Stage B asks whether a source shortlist excludes candidates that remain near the target best. For physical unit i and site c, $Y_{i,c}$ is published annual final yield (hours) for the 11 exact modules moved from Cocoa to Eugene [1]. With positive site-best yield, define percentage deficit:

$$\delta_{i,c} = 100\left(1 - \frac{Y_{i,c}}{Y_c^*}\right), \qquad Y_c^* = \max_i Y_{i,c}, \tag{10}$$

Under Eq. (10), a deficit of zero marks a site-best unit (including ties); a higher percentage denotes a larger shortfall. At a declared nonnegative resolution q in percentage points, retain units whose deficit does not exceed the cutoff:

$$A_c(q) = \{i : \delta_{i,c} \le q\}. \tag{11}$$

Here s and t denote the source and target sites, respectively. A source-to-target false exclusion occurs for any $i \in A_t(q) \setminus A_s(q)$. For an individual exact unit with $\delta_{i,t} < \delta_{i,s}$, the complete interval of resolutions producing that false exclusion is

$$q \in [\delta_{i,t}, \delta_{i,s}). \tag{12}$$

The lower endpoint of Eq. (12) is included and the upper excluded because equality is retained in Eq. (11). Deficit and q are percentages, *not* confidence levels or uncertainty bounds. A numerically chosen q remains a decision resolution even when it happens to equal the approximate comparison-uncertainty percentage reported by the source paper. The same logic applies separately to normalized direct-current (DC) yield in Oslo, Tozeur and Sevilla for matched configurations, not identical physical units.

2.6 Why the two stages are not collapsed into one score

Acquisition excess and shortlist false exclusion describe different harms. Extra characterization increases the evidence burden but does not remove a candidate. False exclusion removes a target-relevant option before target evidence has ruled it out. A composite index would require unobserved trade-off weights and could hide the direction of failure. The stages are therefore reported side by side, with portability considered for the decision stage being reused.

Why symmetric climate distance is not a directional certificate. If $d(s,t) = d(t,s)$, a rule depending only on d must assign the same value to both directions. It therefore cannot exactly reconstruct unequal sequence-replay burdens $E^{\mathrm{acq}}_{s\to t} \neq E^{\mathrm{acq}}_{t\to s}$. Such a distance may still describe pairwise difficulty; the short algebraic argument is retained in the Supplementary Material.

3 Data and evidence boundaries

The evidence sources have distinct roles: published response matrices and measured operating-state conditions define Stage A, whereas observed field yields define Stage B. External operating-state records supply target weights, not additional measurements of the original response-panel products. Table 1 separates the observed evidence from the endpoint that each stage can support. The study does not contain a physical external-site validation of the nine Stage-A response products.

Table 1: Three distinct cohort/endpoint definitions; external operating states weight response profiles but are not new power measurements of those products.

Branch and evidence	What is observed or replayed	Supported outcome and excluded inference
Stage A, fixed response panel	Nine previously published 27-node product-response matrices; source/target weights	Conditional stopping-depth and pair-sign audit; *not* new field product testing
Stage A, external field states	Measured/processed G_{POA}–T_{mod} trajectories from external sites	Target-state support and order replay; *not* nine-product power-label validation
Stage A, matched ten-unit sensitivity	Ten test modules, 18 corrected-power nodes, plus ten distinct Golden calibration units	Fixed-bank modeled test-condition depth; not an IEC time-saving or prospective risk validation
Stage B, exact-unit panel	Published annual yields of the same 11 physical units at Cocoa and Eugene	Threshold-conditional shortlist exclusion; *not* an uncertainty-calibrated decision
Stage B, matched configurations	Normalized DC observations from Oslo, Tozeur and Sevilla	Configuration-level candidate-set contrast; *not* identical serial-unit transfer

3.1 Operating-state distributions and response panel

The characterization-order experiment uses the U.S. National Renewable Energy Laboratory (NREL) mPERT operating-state records from Cocoa, Florida; Eugene, Oregon; and Golden, Colorado; one year of five-minute IEA PVPS Task 13 measurements from Switzerland; and the Sandia PV Module Operating Temperature dataset from Albuquerque, New Mexico [27–30]. Across the 15 NREL core files, 462,429 quality-filtered records have finite timestamps, plane-of-array irradiance, module temperature and maximum-power quantities. Site weights average five technology-specific field trajectories. Switzerland contributes 36,449 five-minute rows; 34,763 pass the declared validity screen and 28,292 (81.39%) lie inside the retained response-grid triangulation. Albuquerque contributes 49,939 valid five-minute daylight bins, of which 36,911 (73.91%) lie inside the triangulation. States outside supported geometry are excluded rather than extrapolated.

3.2 Exact-unit and matched-configuration field panels

The shortlist experiment begins with the complete Cocoa–Eugene annual final-yield panel published by Marion et al. [1]. The same 11 physical modules were moved from Cocoa to Eugene, allowing an identity-controlled transfer test, but not a simultaneous cross-site experiment. The published annual endpoints cover March 2011–February 2012 for Cocoa and January–December 2013 for Eugene [1]. The accompanying NREL source manual records full deployments from 21 January 2011 to 4 March 2012 (Cocoa) and from 20 December 2012

to 20 January 2014 (Eugene), together with post-deployment STC checks and time-varying performance diagnostics [28]. Identical unit identity therefore does not control calendar period, deployment history or time-dependent module response; the reanalysis does not isolate a causal climate effect. Golden used a second set of physical units from the same manufacturer/model families; those units do not enter the eleven-unit field shortlist, but ten complete laboratory profiles form a separate reference bank for the supplementary sequence audit. Marion et al. report an expanded relative uncertainty of 3.5% for annual final PV yield Y_f at coverage factor $k = 2$, and an uncertainty of approximately 5% when comparing two annual yields using their stated combination rule (Table 3 and Section 3) [1]. These published values are acknowledged as measurement context, not transformed into intervals for the normalized deficits or target-set memberships below.

A separate contemporary field comparison uses the public Zenodo archive released with the 2026 Solar Energy multi-climate study of nominally identical panel configurations in Oslo, Tozeur and Sevilla [2, 31]. Raw module-level Tigo observations are minute-spaced in the verified archive and are aggregated to five-minute means without interpolation. Module power is normalized by public STC rating; a configuration is admitted when at least 75% of assigned modules have positive finite values, and a site timestamp is retained only when all seven configurations are defined. The primary endpoint integrates 23 common calendar months across the three sites. Fifty- and 100%-coverage rules, two temporal blocks and paired-month bootstrap resampling are retained as robustness checks. This is configuration-level evidence, not exact-unit redeployment.

3.3 External operating-state targets and numerical admission

To test out-of-source Stage-A portability without retuning the original nine product-response matrices, the five frozen historical node orders are applied to four independently documented field-data families. In Safi, Morocco, original five-minute plane-of-array irradiance is synchronized within two seconds to each of three measured module-temperature series (m-Si, p-Si and a-Si); these trajectories cover unequal periods at *one* location [32, 33]. MetPVNet contributes three one-minute-resampled German PV-system campaign files from two nearby stations in the Allgäu region, not three independent climate types [34, 35]. A nine-year, 15-minute-mean field series from one Trieste installation provides one primary module-temperature sensor with two alternative same-installation sensors and calendar-year sensitivity checks [36]. BR-PVGen contributes 51 anonymized Brazilian plant weather-station files, whose 15-minute measurements were published after weighted-moving-average processing and selective gap interpolation [37].

For the Brazilian primary screen, both sensor-level source-sample counts must be positive; an explicit interpolation-TRUE flag on either irradiance or module temperature rejects that row. A missing flag is not evidence that no aggregation occurred. Each eligible plant requires at least 1,000 simultaneous positive-G_{POA} observations within the frozen 27-node interpolation grid and at least 30 distinct observed dates; 34 of 51 meet this numerical gate. The same minimum is applied to German and Italian trajectories. No unsupported state is extrapolated. The original five-site 25-cell result matrix is reproduced to within 10^{-8} percentage points *before* testing any new target; all five old orders, the 36 held product pairs, empirical strict-stop rule, and target-local reference are unchanged. A source with unresolved pair(s) has no complete-36-pair mean-excess comparison. The observational unit is a single observed field *operating-state distribution*, not a measured power response of any of the nine original benchmark products. Source hashes, licensing and separate sensor/time-span gates are given in the Supplement and executable archive. A separate, source-verified

support-boundary diagnostic was performed without changing the frozen node grid, target weights, stopping rule, or any of the original 25 transfer cells. Across the three German campaigns, the primary Italian sensor, 34 eligible Brazilian plant files, Switzerland, and Albuquerque, all 40 source-specific in-grid record counts reproduce the archived admissions exactly. The out-of-grid records are classified geometrically, with exclusive priority to low irradiance, high irradiance, low module temperature, high module temperature and the sloped convex-hull boundary (Supplementary Table S7). This diagnostic describes *which observed operating states are removed*, not the performance of products under those omitted states. All subsequent depth comparisons therefore target the admitted, renormalized within-grid distribution, not the full field-state climate. Particularly restrictive support is visible in Safi (40.32–44.01% across the three technology trajectories) and Italy (57.4%); the unobserved outside-grid contribution cannot be bounded from these replays. In particular, excluded-state composition must not be interpreted as a causal explanation of directed transfer. The corresponding Safi exclusion breakdown is not inferred from summary counts alone.

4 Results

The findings are presented first by Stage-A decision task and then by the separate Stage-B screening operation. Pair-average, winner-identification, and campaign metrics are computed from the same stopping records but answer different operational questions. Cross-stage comparisons remain interpretive, not pooled product-level tests.

4.1 Acquisition-order transport is strongly directional

All 36 full-reference pairwise signs agree across Cocoa, Eugene and Golden. Switzerland and Albuquerque each agree with Eugene on 35/36 response-panel signs; the one changed pair is reported rather than suppressed. Despite this near-stability of the full comparisons, the complete directed acquisition matrix is highly asymmetric (Fig. 1).

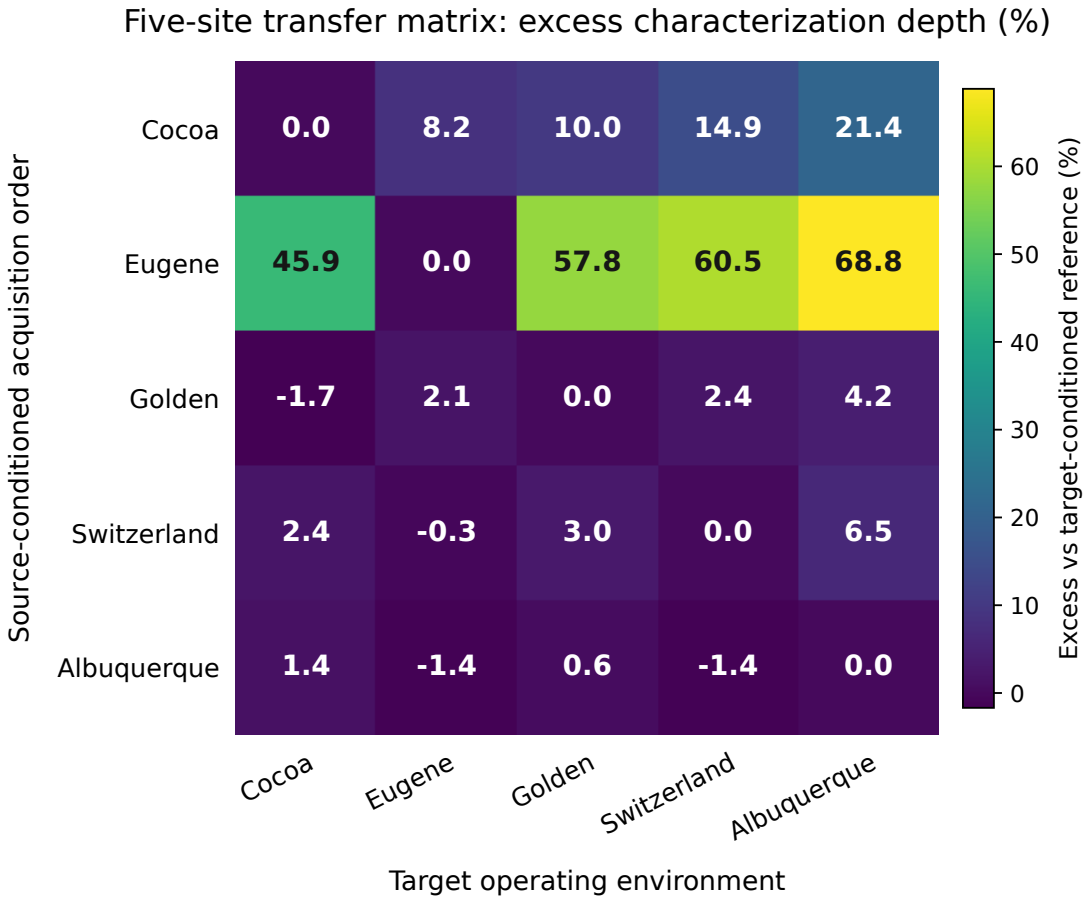


Figure 1: Complete five-site directed characterization-order transfer. Each cell is excess mean stopping depth relative to the target-conditioned reference. All 25 aggregate cells resolve all 36 response-panel comparisons with no wrong accepted stopping decision.

The off-target source summaries are sharply separated. Eugene has +58.25% mean off-target excess and a +68.83% worst target. Albuquerque is -0.19% on average, with all four off-target values between -1.37% and +1.35%; Golden and Switzerland are +1.74% and +2.89%, respectively, and Cocoa +13.64%. The percentage differences correspond to substantial absolute changes in the declared measurement proxy. For Eugene→Albuquerque, mean stopping depth rises from 8.56 target-conditioned cells to 14.44 cells (+5.89 cells; pair-level median extra 5, interquartile range (IQR) 3–9). The reverse Albuquerque→Eugene transfer is 7.97 versus 8.08 cells (-0.11; pair-level median extra 0, IQR 0–1). Likewise, Eugene→Switzerland is 13.19 versus 8.22 cells (+4.97; median extra 4.5, IQR 2–7.25), whereas Switzerland→Eugene is 8.06 versus 8.08 cells (-0.03; median extra 0, IQR 0–1). Thus the directional effect is visible in absolute test cells, not only in ratios: transferring Eugene's order to Albuquerque imposes an appreciably longer retrospective characterization path, while the reverse does not. This remains a count of model-evaluated conditions, not observed measurement minutes.

A prespecified Switzerland-as-target reverse check retains the response panel, interpolation and stopping rule: Albuquerque→Switzerland is -1.35% and Golden→Switzerland +2.36%, versus +60.47% for Eugene→Switzerland. Under the common supported 27-simplex subset, Albuquerque remains at -1.00% mean off-target excess (worst 0.00%).

Symmetric Jensen–Shannon, Hellinger, total-variation and normalized two-dimensional Wasserstein distances are descriptively associated with mean bidirectional burden (Spearman approximately 0.79–0.85 over ten environment pairs), but they cannot explain direction

because one distance value represents both transfer directions. Thus state-distribution mismatch helps describe pair difficulty, while portability remains an order-specific directional property.

4.2 External operating-state replay: distinct sources remain context dependent

Table 2 reports relative measurement depth when each fixed historical sequence is applied to an observed G_{POA}–T_{mod} target. The unchanged reference program reproduces all 25 original transfer-matrix cells exactly. Morocco contains 42,332, 13,197 and 26,194 within-grid raw paired records for m-Si, p-Si and a-Si, respectively. Eugene has the lowest excess among the five *historical* orders on each of those three targets (+6.47%, +12.70%, +8.66%); its rank is not invariant to source-provided preprocessing, which makes Albuquerque slightly less burdensome for the m-Si subset. The raw time-alignment and small full-pair margins therefore remain visible sensitivity limits.

Germany's two autumn-2018 and one summer-2019 system campaigns have 19,937, 19,800 and 21,069 retained within-grid observations. Albuquerque has the lowest burden among the historical source orders for both autumn files, while Golden has the lowest for summer 2019. The latter target produces one wrong accepted *near-tied* product comparison under each of Cocoa, Golden, Switzerland and Albuquerque orders; the full normalized pair margin is about -5.3×10^{-5}. This is a model-rule sign discrepancy, not an identified physical module-power difference. For Italy, the primary module-temperature series retains 86,859 of 151,211 paired positive-POA records across 2013–2021. Switzerland, Albuquerque and Golden all require fewer cells than the target-conditioned heuristic on the full period, with Switzerland the smallest by a narrow margin; alternate sensors and calendar-year splits show additional within-site variation.

Brazil has 398,394 paired, source-sample-supported daytime 15-minute records across 34 eligible plants, with 269,858 inside the unchanged response grid. Across these 34 *plant-level* target distributions, median acquisition excess is -3.31% for Albuquerque, -2.83% for Golden, -0.17% for Switzerland, +1.38% for Cocoa and +39.69% for Eugene, whose median uses only 32 complete-36-pair plants. The Eugene order leaves three comparisons unresolved in the other two plants; there are no wrong accepted signs on resolved pairs in the 34 full-period plant tests. These results have broad plant-to-plant dispersion (Fig. 2) rather than a globally stable source hierarchy. In the separate, dependent calendar-quarter sensitivity, 30 of 31 plants with two or more eligible quarterly windows have no source order simultaneously lowest in every window, counting tied minima as shared (Supplementary Fig. 2). This is a descriptive time-window contrast; quarters from one plant are *not* independent sites and no confidence limit for a population of climates is inferred.

Table 3 reports the full decision-status denominator rather than reporting only mean stopping depth; the preceding Table 2 reports the separate target support fractions, since a count of order replays cannot stand in for climate coverage. The 190 historical-order runs across 38 primary external target distributions entail $190 \times 36 = 6{,}840$ held comparisons: four wrong accepted signs, all for the same near-tied German pair ($|\Delta| \approx 5.3 \times 10^{-5}$), and three unresolved pairs under the Eugene order in two Brazilian plant records. Thus 188 of 190 runs are fully resolved, but complete resolution *alone* does not imply sign correctness. These counts are independently re-aggregated from archived pair-level replay outputs; they are not extra measurements, an estimated population error rate or a calibrated confidence bound.

Table 2: Stage-A external replay and in-grid support. The five numeric order columns give excess relative characterization depth (%). The "In-grid" column is the share of valid paired daylight G_{POA}–T_{mod} records retained by the fixed response geometry, not the share of physical-module measurements validated. Negative excess means fewer modeled test cells relative to the target heuristic, not optimality. Brazil reports medians of 34 plant-level excesses but its in-grid fraction is pooled over eligible plant records; these are different aggregation units.

Observed target	In-grid (%)	Cocoa	Eugene	Golden	Switzerland	Albuquerque
Safi m-Si	42.86	+19.83	+6.47	+12.93	+13.79	+8.62
Safi p-Si	40.32	+51.23	+12.70	+40.57	+27.46	+29.10
Safi a-Si	44.01	+26.84	+8.66	+19.05	+16.88	+13.42
Germany 2018/S1	65.1	+9.36	+15.74	+1.70	-0.85	-3.83
Germany 2018/S2A	69.9	+15.32	+18.72	+3.83	+2.98	-0.43
Germany 2019/S2B	65.1	+8.65	+3.76	+3.01	+4.89	+3.76
Italy Q2/inv1	57.4	+4.86	+19.79	-2.78	-3.47	-3.12
Brazil plant median[†]	67.7	+1.38	+39.69	-2.83	-0.17	-3.31

[†] For Eugene, 32/34 plant means are fully resolved; the other two have three unresolved pairs collectively. The other four orders resolve all 34 plant records. Safi's three technology subsets are from one site. In-grid retention is conditional on each source's paired validity gate; excluded states have no evaluated response-panel outcomes, so results cannot be generalized to the full climate without new measurements; see Supplementary Tables S4 and S7.

Table 3: Three-state audit of the primary Stage-A replay outputs: 38 field-state targets, five historical source sequences per target. Wrong accepted and unresolved counts refer to held product-pair decisions, *not* independent physical module measurements. Morocco's separately admitted trajectories and correlated sensitivity windows are outside this denominator.

Input family	Targets	Runs	Wrong	Unresolved
Germany	3	15	4	0
Italy	1	5	0	0
Brazil	34	170	0	3
Total	38	190	4	3

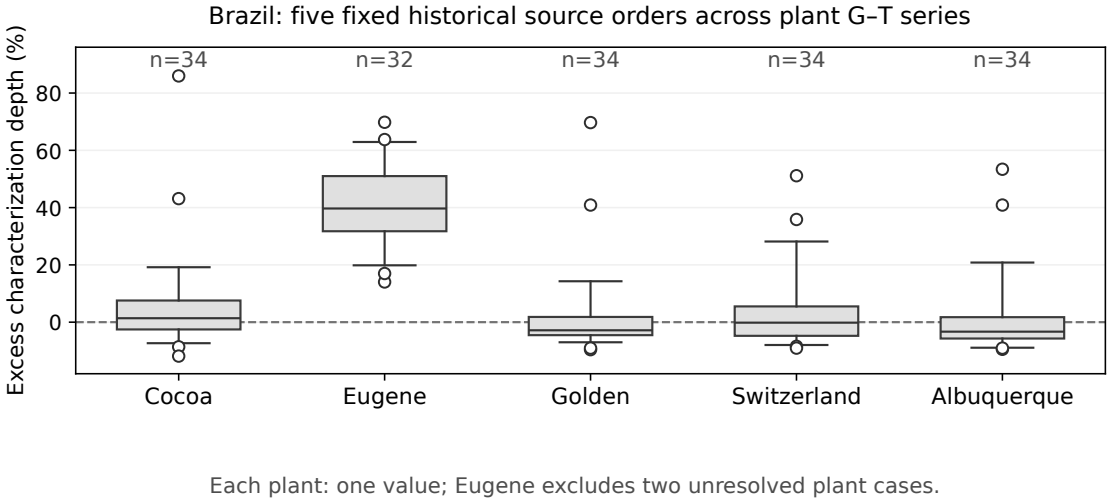


Figure 2: Brazilian field-state distributions and frozen-source acquisition excess. Each of 34 eligible plant files contributes one value per historical source order when all 36 comparisons resolve; Eugene is omitted from the box statistics for two sites with unresolved pairs. The figure describes this public, anonymized plant corpus and does not estimate an independent climate-population distribution.

4.3 Portability changes with the decision task

The original nine-product Cocoa–Eugene result already shows a direct objective reversal. Relative to Eugene's target-conditioned heuristic, the Cocoa order raises mean pairwise depth from 8.08 to 8.75 cells (+8.25%), raises K_{all} from 14 to 17, and raises $N_{\text{mod}-\text{cell}}$ from 118 to 133. Yet the same Cocoa order produces an accepted winner that matches the full-reference winner at depth 8 rather than 14 (-42.9%). The Golden order behaves similarly on Eugene: its mean is slightly worse (8.25 versus 8.08), while winner depth is again 8. Thus a sequence judged nonportable for complete pair resolution can be favorable for winner identification on the same response panel. Conversely, Eugene→Cocoa is worse under both pair-average and winner-depth tasks.

The external replay confirms that this is not only a single core-panel construction. All 190 historical-source runs on the 38 primary external target distributions produce a unique accepted winner; four are incorrect relative to the full reference, all for the same near-tied German comparison already identified in the pairwise audit. Of the 186 correct accepted winners, two Eugene-order Brazilian runs still contain unresolved non-winner pairs and therefore have no complete all-pairs mean. Restricting the cross-task comparison to the remaining 184 correct, all-pairs-complete runs, Table 4 compares the direction of source-versus-local change in pairwise mean and winner depth. Twelve runs show a strict reversal: nine have worse pairwise mean but better winner depth, and three have better pairwise mean but worse winner depth. Two additional runs have unchanged pairwise mean but different winner depth. For transparency, an efficiency-only comparison ignoring winner correctness would include 188 complete all-pairs runs and yield 16 apparent strict reversals. The four additional worse-mean/better-winner-depth cases *all identify an incorrect winner* at the same German 2019/System 2B target (Cocoa, Golden, Switzerland and Albuquerque orders). They cannot be counted as valid winner-identification improvements; the 184/12 subset applies an explicit *retrospective validity gate*, not a prospectively observable certificate. These runs are correlated reuses of the same nine response matrices and are reported as finite cases, not as independent sampling frequencies. The 12 valid reversals occur on 11

distinct target records, with nine run-level cases from the Brazilian plant-data family. Nine changes in accepted-winner depth are only one test cell in magnitude, and every absolute pairwise-mean change is below 5%. These magnitudes limit the interpretation: strict task disagreement is established in the archived replay, but the external case count is neither 12 independent climate trials nor evidence of a large typical operational effect (Supplementary Table S14).

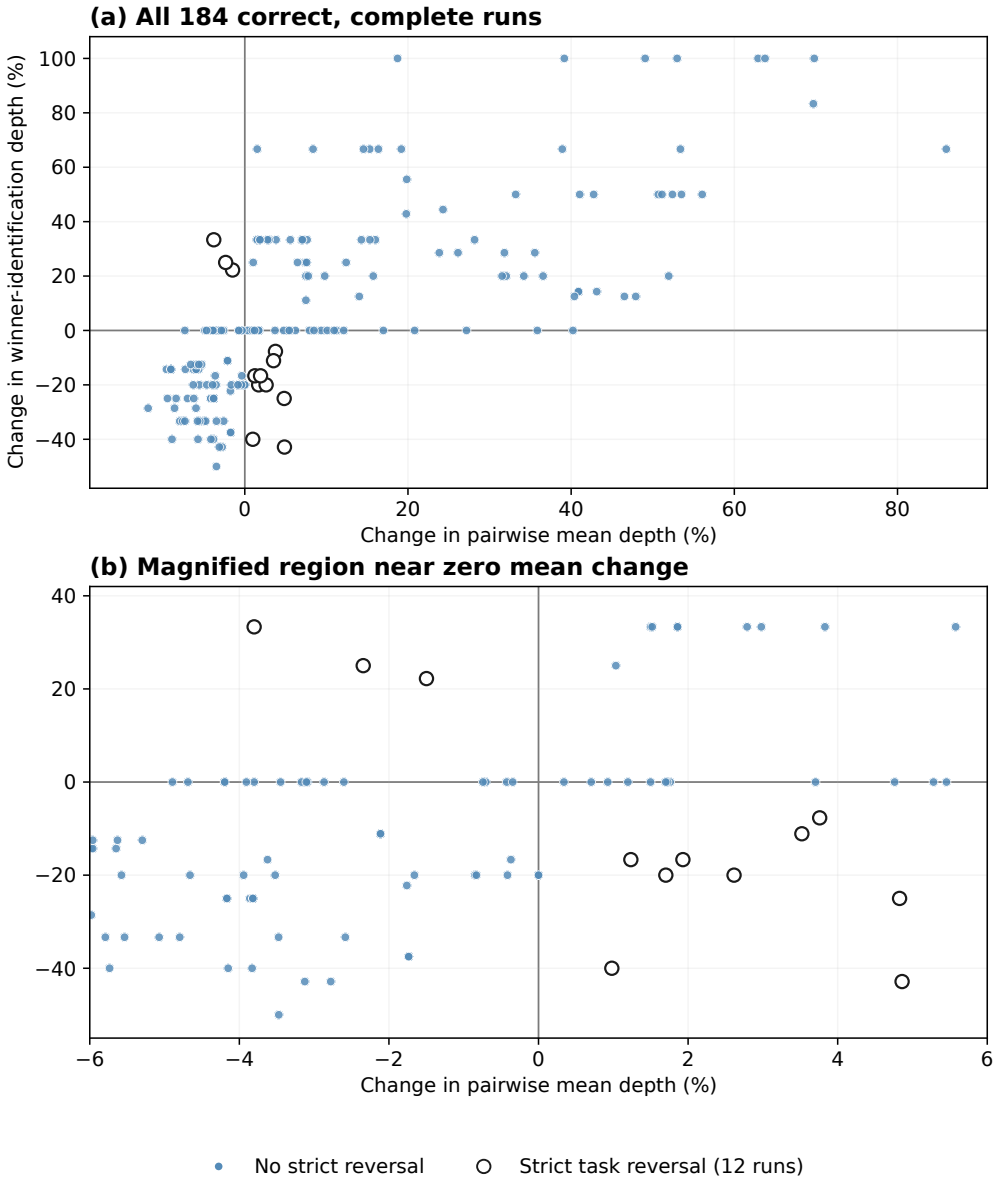


Figure 3: Task-conditioned portability on 184 primary historical-order runs with a retrospectively correct accepted winner and complete all-pairs stopping depths. (a) Full extent; (b) magnified view of the same records near zero pair-mean change. Open circles mark 12 strict sign reversals; no observations are interpolated, moved, or resampled. Runs share a response panel and correlated target records, so counts are descriptive rather than a population-frequency estimate.

Table 4: Task discordance for the same 184 correct, all-pairs-complete runs in Fig. 3. Entries count source orders by whether they are better, equal, or worse than the target-conditioned heuristic.

Pairwise mean \ winner depth	Better	Same	Worse
Better	59	17	3
Same	2	0	0
Worse	9	25	69

A shared all-pairs campaign can disagree with the pairwise mean even when winner identification does not. Among 79 fully resolved historical-order runs with a lower mean than their target-local reference, 26 increase $N_{\mathrm{mod-cell}}$, 40 increase K_{all}, and 21 increase *both*. Those 21 remain worse under any nonnegative weighted sum of the two campaign proxies with at least one positive weight. For Brazilian record `D09_PS_001`, Albuquerque improves the mean 7.92→7.17, but increases K_{all} 14→15 and $N_{\mathrm{mod-cell}}$ 100→102; its winner depth simultaneously improves by two cells. A lower pairwise mean therefore does not identify the preferred sequence until the decision task is specified.

All workload proxies exclude the initial calibration-profile acquisition as well as physical setup, stabilization and scan time; none establishes a net laboratory-time saving.

4.4 Restricted fixed-Golden sensitivity: task agreement rather than reversal

The restricted ten-unit fixed-Golden-bank experiment provides a counterpoint to the task reversals above. Cocoa→Eugene increases mean modeled depth by 27.49%, K_{all} from 13 to 14, $N_{\mathrm{mod-cell}}$ from 122 to 138, and accepted-winner depth from 9 to 13. Eugene→Cocoa increases the corresponding quantities by 29.10%, 16 to 17, 132 to 151, and 10 to 12. Thus task discordance is not universal across cohorts: the reusable conclusion is that portability must be indexed by task, not that winner identification is generally easier to transfer.

With the fixed Golden bank, Cocoa→Eugene has 42/45 pairwise depth penalties and three ties, while Eugene→Cocoa has 45/45 penalties. All 180 fixed-bank decisions resolve and their accepted signs agree with the retrospective complete laboratory contrasts. Across all 56 overlapping eight-to-ten-Golden-unit calibration subsets, mean directional penalties remain positive but range from +5.51% to +30.35% (Cocoa→Eugene) and +26.55% to +31.54% (reverse); these 10,080 accepted-sign checks are dependent replays, not separate physical trials (Supplementary Tables S8–S10).

More critically, two held pairs under the full bank and Cocoa→Eugene have actual residuals exceeding the bank maximum at their stopping prefix: `CIGS39017/mSi460A8` (0.00370685 versus 0.00358310) and `aSiTandem72-46/xSi12922` (8.63379×10^{-5} versus 8.01512×10^{-5}). Their signs happen to agree with the full reference, but the bank maximum is demonstrably *not* a deterministic unseen-pair error bound. The restricted 18-node grid admits 71.07% of qualifying Cocoa states and 57.50% of Eugene states. No full-climate or prospective stopping guarantee follows.

Random-permutation and one-swap comparators are reported in Supplementary Table S11. They use completed held-pair outcomes and are retrospective diagnostics, not independently tested or deployable sequence-design rules.

4.5 Secondary extension: portability of candidate shortlists

Stage B asks a different set-valued question and is retained as a secondary extension rather than a second coequal contribution. For the same 11 physical modules operated first in Cocoa and later in Eugene, the source and target retain different candidates even though the calculation uses the same nominal deficit definition. At $q = 5\%$, the Cocoa list excludes three of the five Eugene-retained units. The stronger result is interval-based: HIT05667 is excluded by Cocoa but retained in Eugene for every $q \in [3.239, 8.123)\%$, and the union of exact-unit intervals contains at least one Cocoa→Eugene false exclusion continuously over $[3.104, 13.819)\%$. The reverse direction has no exact-unit false-exclusion interval over the declared 0–16% sweep. Full deficits, interval endpoints and the q sweep are reported in Supplementary Sections 7–9.

This is a nominal point-estimate diagnostic, not an uncertainty-calibrated screening certificate. One of the three $q = 5\%$ Cocoa→Eugene exclusions lies only 0.052 percentage points beyond the source cutoff, and the source data do not provide the joint covariance model required to propagate uncertainty through the site-best normalization and selected maximum. A separate Oslo–Tozeur–Sevilla matched-configuration archive supplies a qualitative replication of the set-valued phenomenon: M1-HFL remains the point leader at all three sites while M1-Bi crosses the 5% candidate boundary in both warmer targets; its exact directional intervals and temporal checks remain in the Supplement. Neither archive estimates a climate-only causal effect.

4.6 Relation between sequence and shortlist diagnostics

The two diagnostics should not be collapsed into one score. Sequence reuse concerns how much of a fixed measurement path is needed for a declared decision task; shortlist reuse concerns whether a source-conditioned candidate set omits target-relevant alternatives. They use different physical cohorts and endpoints in the primary analyses. The common lesson is narrower: final-answer stability is insufficient evidence that the *path used to reach or simplify that answer* is portable. No cross-stage cost–exclusion trade-off or joint operating policy is estimated.

5 Robustness and common-sequence comparators

Acquisition asymmetry persists under five-site common-simplex support, 1,000 matched random orders per off-target direction, pooled and target-excluded common orders, and technology-stratified U.S. replays (Supplement). The equal-site pooled order has +5.61% mean/+11.36% worst-target excess; the target-excluded LOCO5 order has +13.77%/+25.65%. Neither is a proven optimum. One Eugene-HIT trajectory contains a wrong accepted sign for several otherwise low-burden orders, precluding a universal sign-preservation claim.

Stage-B false exclusion is checked across all q, exact-unit intervals, coverage thresholds (50/75/100%), two temporal blocks and paired-month bootstrap samples. The external reconstructed M1 bifacial–monofacial contrasts (9.995%, 21.464%, 8.949%) closely match the published approximate 10/21/8% values [2]; this is a processing check, not new product performance evidence. Germany's campaigns, Italy's repeated sensors/years and Brazil's correlated plant/quarter windows are distinct dependency structures, not an enlarged sample of independent climates. Short-window wrong signs and changing source preferences reinforce the empirical, non-calibrated status of the stopping rule.

6 Discussion

Together, the results distinguish transport of PV evidence from transport of module performance. Recent climate-specific PV studies emphasize that module design, degradation and energy yield depend on local stressors and operating conditions [2–5]. This study shows that *the evidence pathway itself* has a transport problem distinct from module performance transport. A source environment can yield a measurement order that is inefficient elsewhere or a shortlist that is too restrictive, even when the eventual ranking appears stable.

Source selection by original five-site averages would favor Albuquerque, whereas the frozen source orders favor Eugene for the Moroccan raw-data trajectories, with different preferences in Germany, Italy, and Brazil. The admissible unit is a *source order and target support/period pair*, not a universal country ranking. Brazilian quarterly contrasts are dependent time-window diagnostics, not causal seasonality evidence.

6.1 Support-limited operating states

The fixed 27-node hull excludes low-irradiance and other unsupported states; the original-source diagnostic finds low irradiance the largest geometric exclusion category in Germany, Italy, and Brazil (Supplementary Table S7). Renormalizing the retained target weights therefore estimates transfer burden for a *conditional supported distribution*, not the complete climate. With only 40–44% retained in Safi or 57.4% in Italy, the full-distribution contrast is unidentified.

For planning, the auditable output is a *candidate sequence plus supported-state fraction and pair-decision record*. Insufficient support or unresolved pairs prevents a complete 36-pair comparison. A pooled sequence may reduce worst-target burden against a poor single-source order, but the evaluated pooled and full-campaign minima are hindsight diagnostics, not deployable optima. Actual instrument time or cost requires acquisition and stabilization logs.

6.2 Retrospective use with an existing calibration library

The source-order experiment is best interpreted as an *audit of fixed-protocol reuse*, not as a recommendation to ignore target information. If target operating-state weights are already available and a laboratory is free to redesign its sequence, there is no general reason to prefer a historical source order; even the target-weight-descending order is only a heuristic and need not minimize winner depth or full-campaign burden. The transport question is scientifically relevant when a sequence is fixed before target-specific redesign, retained for cross-site comparability, or audited after deployment. The analysis therefore asks what changes when a pre-existing sequence is reused, not whether a rational unconstrained optimizer should choose it. The rotating seven-profile calibration sets presuppose the complete nine-product reference archive. The restricted fixed-Golden-bank test avoids the tested unit identities but still presupposes 180 calibration observations from related model families. Neither $\bar{k}$, K_{all}, nor $N_{\mathrm{mod-cell}}$ includes library acquisition and maintenance. Thus neither configuration establishes net savings or a validated decision rule for unseen products; Table 5 states the distinct prerequisites.

Table 5: Operational prerequisites and scope of inference. No laboratory time, monetary saving or newly tested-product error rate was measured.

Use case	Evidence that must already exist	Permitted interpretation and unresolved burden
Retrospective nine-product archive	All nine complete response profiles; seven 27-node calibration profiles for each held pair; supported target-state weights	Replicates the tested comparison/depth audit; acquisition and maintenance of the archive are not counted.
Prospective new product pair	Existing independently relevant reference profiles, target-state observations and newly measured product responses at visited conditions	A potential workflow to validate, *not* a demonstrated risk-controlled stopping policy on unseen products.
No reference library	Full response acquisition and calibration, plus setup, stabilization and repeat scans, would be needed before the retrospective audit	Net test time and cost cannot be inferred and no payback or laboratory-time reduction is established.

For a laboratory with a pre-existing, relevant reference library, the audit can compare fixed orderings for the *declared* purpose: all-pairs completion or accepted-winner identification. Because correctness is assessed against the complete archived response only after the replay, a short accepted-winner depth cannot be treated as permission to stop a new physical campaign without a separately validated uncertainty and error-control procedure.

The distinction between a model's stopping rule and a genuine measurement-based decision rule is fundamental: Ferrero and Petri analyze how measurement uncertainty and incorrect-decision risk enter threshold decisions [38]. The residual envelope used here is estimated from a fixed calibration-product panel, not from traceable standard uncertainties; consequently, neither the four observed near-tie discrepancies nor the absence of other accepted errors can be converted to a calibrated error probability. This separation is central to interpreting the Stage-A test-cell proxy.

Recent measurement studies provide a different uncertainty perspective: Herrmann et al. report expanded CSER uncertainty near ± 2.0–$\pm 2.3\%$ [16]; Lyubenova et al. demonstrate sensitivity of outdoor G–T matrix reconstruction to sampling duration and season [14]; and Huang et al. evaluate repeatability of STC-corrected power under dynamic field conditions [39]. None supplies cell-level bounds for this archived response panel. Their published uncertainty percentages cannot be imported as empirical stopping-risk guarantees.

For prequalification, the relevant test is whether the source shortlist preserves target-relevant alternatives at a stated resolution, not merely whether its leader remains unchanged. This aligns with recent work on distributionally robust feature selection, where a limited evidence subset is chosen to remain useful across multiple populations, and with safe-screening theory that treats false elimination as a distinct decision error [23, 24]. The present work is different in scope: it provides empirical PV field counterexamples rather than a formal distributional guarantee. Unlike the calibrated energy-rating cost study of Blakesley et al. [13], the current records support a *direction-dependent diagnostic* of reusing fixed measurement orders and candidate lists. The 21 full-campaign proxy reversals show why lower mean pairwise depth alone cannot establish reduced laboratory burden; the analysis does not offer an alternative uncertainty-calibrated reduced IEC measurement schedule.

The 2026 IEA PVPS project-decision framework emphasizes the links between component selection, monitoring, quality and downstream value across the PV lifecycle [6]. This distinction suggests a refinement to evidence provenance: records should identify not only

where a measurement came from, but also *which decision stage it is intended to support after transfer*. Climate similarity, a stable winner and a good source-site fit are all insufficient substitutes for a stage-specific transport evaluation.

7 Limitations

Stage A combines a fixed nine-product response panel with field operating-state weights from *different modules*; the external sites do not measure those nine products. The empirical stopping envelope is neither a metrological uncertainty budget nor a prospective safety certificate, and target-conditioned orders are heuristics, not global optima. Cell-specific response uncertainties are unavailable, so no arbitrary perturbation bounds are imposed. Switzerland and Albuquerque each contribute one field trajectory whereas each U.S. core site averages five; Albuquerque results condition on the 73.91% of valid daylight states inside the response triangulation, with no extrapolation beyond it.

The extended evidence is likewise source- and support-conditioned. Three Moroccan technology trajectories share a location but cover unequal periods; Germany has short campaigns at two nearby stations; Trieste has repeated sensors and annual subsets from one array. Brazil's plant locations are anonymized to state, systems may share conditions, and source-published 15-minute readings include moving-average processing and selective interpolation. Rows explicitly flagged as interpolated were excluded, and positive source counts were required; however, absence of a flag does not certify raw five-minute data. Its 34 admitted plant records are not 34 independent climates. Near-tied signs, strict stopping and preprocessing remain uncalibrated limitations, and external Stage-A sources supply *no new Stage-B false-exclusion tests*.

The primary two stages share the Cocoa–Eugene climate names but *not* a common physical module cohort: Stage A holds nine response matrices fixed, while Stage B evaluates 11 redeployed field units. The secondary ten-unit sensitivity shares the same test-unit identities across its laboratory and field calculations and uses a fixed bank of separate Golden physical units. Both groups were characterized in the historical laboratory archive, using the same source family of manufacturer/model profiles; laboratory and field endpoints are noncontemporaneous. It is not a simultaneously measured prospective trial. The contrast across stages is not within-product causal or operational validation. At $q = 5\%$, one of the three Cocoa→Eugene nominal false exclusions is only 0.052 percentage points beyond the source cutoff. The original source reports an expanded annual-Y_f uncertainty and an approximate two-yield comparison uncertainty, but does not supply the joint, site-best-normalized uncertainty information needed here to certify candidate membership or exclusion as metrologically robust. The three-out-of-five result must not be interpreted as three instrument-confirmed omissions. The original measurements also compare the same units over different calendar years, with NREL documenting module-performance and spectral-history considerations and post-deployment checks [1, 28]. Unit identity removes a cross-product identity difference, but does not identify a climate-only causal contrast or rule out time-varying module effects. The exact-unit Cocoa–Eugene panel uses historical modules and is not a procurement statement about present TOPCon, HJT or bills of materials. Oslo–Tozeur–Sevilla provides matched configurations rather than a redeployment of identical physical units; its normalized DC endpoint is not pooled with the former annual final-yield endpoint. Neither stage establishes a causal weather mechanism, monetary benefit, minimum IEC-required test subset, universal shortlist or calibrated decision-risk bound. Instrument histories, cell-level reference uncertainties and stabilization-time logs needed for those claims are absent.

Future uncertainty-aware analysis would require measurement-calibrated joint bounds on node weights and response values and an independent, matched-product field or laboratory campaign. The required calibration records are unavailable; no uncertainty-calibrated stopping certificate or prospective safety guarantee is claimed.

8 Conclusion

PV characterization evidence is not automatically portable merely because the preferred product or pairwise full-reference signs remain stable, and portability cannot be summarized by one stopping-depth statistic. In the same nine-product Cocoa–Eugene panel, a transferred sequence is worse for mean all-pairs resolution yet substantially better for accepted-winner identification. Across external targets, strict task reversals recur, while the fixed-Golden ten-unit sensitivity shows that they need not occur in every cohort. The algebra of the pair-depth graph explains why: edge-average, edge-maximum, vertex-maximum sum, and winner-star depth are different functionals of the same stopping records.

Measurement-sequence reuse should therefore be reported *by decision task*: pairwise comparison, winner identification, or completion of a shared characterization campaign. Candidate-shortlist reuse remains a separate directional screening problem whose exact threshold intervals reveal where target-relevant units are excluded. External Stage-A replays expose target and time-window dependence, four wrong accepted near-tie decisions in one German pair, and three unresolved decisions in two Brazilian records; the Golden sensitivity additionally shows that its empirical envelope is not a deterministic bound. These are evidence-bounded diagnostics, not new nine-product field measurements, calibrated error guarantees, universal safe test reductions, or demonstrated laboratory-time savings. The reusable lesson is to declare the decision task first and then audit whether the evidence path remains suitable at the target.

Supplementary Materials

Supplementary Methods and Results are provided in a separate PDF file.

Author Contributions

Ghassan Malkawi: Writing—original draft; manuscript preparation and development. Ahmed Elsayed: Writing—original draft; manuscript text development, and review of the analysis and verification code he worked on. Both authors: Writing—review and editing; responsibility for scientific integrity.

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Institutional Review

The analysis uses previously published photovoltaic and environmental datasets and does not involve human or animal participants. Human-subject ethics approval and participant consent are not applicable.

Informed Consent

Not applicable.

Data and Code Availability

The underlying PV and operating-state records are identified by source DOIs or URLs and file hashes where available. They comprise NREL mPERT, IEA PVPS Task 13 Switzerland, Sandia Albuquerque, published Cocoa–Eugene annual yields, and Oslo–Tozeur–Sevilla matched-configuration observations. External Stage-A sources are the Safi workbooks, MetPVNet Germany, Trieste Q2 and BR-PVGen [33, 35–37]. Original producer spreadsheets and raw logger observations are not redistributed.

The public GitHub companion at https://github.com/shaikhamalkawi-ux/pv-evidence-portability-across-climates provides recovered original Stage-A programs, derived-output verifiers and exact-unit Stage-B shortlist checks (historical checked snapshot `9a8dea54`). The publicly downloadable R18 validity-gated comparison ledger, its two standard-library verification scripts, the 12-case Table S14 source, source hashes and a display-only figure replot are frozen in the `cb42962c` R18 reproducibility directory; the figure code and vector PDF share the same commit. These derived records enable direct replay of the 190/188/184/12 census but do not contain third-party raw I–V, performance or meteorological observations. The author-controlled replay archive preserves program and input hashes, target admissions, five-order excess depth and all 36 pair-level decisions per target. The recorded source-file replay regenerated 900 order–target rows and 32,400 pair decisions exactly; separate output re-aggregation passed 26/26 checks (Supplement). These are computational checks, not independent physical validation.

The accompanying reviewer reproducibility files provide two additional routes. The Table S7 support-audit bundle contains the source-audit script, frozen grid and admission registry, derived CSVs, input hashes and instructions for obtaining the required producer files. The matched-ten toolkit uses the original NREL laboratory workbook, an 18-node grid, ten physically distinct Golden calibrators and frozen site-node weights. It includes offline replay of the fixed-bank experiment, 56 bank subsets, 5,000 seeded random orders and 136 single swaps per destination, with reference outputs, separate implementation checks and a nominal shortlist calculator. This route is conditional on the archived weights; it does not regenerate raw meteorological weights or re-execute the original multicountry and external field-yield analyses. These additional files are supplied for review and are not represented as already available in the public GitHub snapshot. Reproduction requires the corresponding original inputs under their source terms.

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Conflicts of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this manuscript.

References

[1] W. Marion, M. G. Deceglie, and T. J. Silverman, "Analysis of measured photovoltaic module performance for Florida, Oregon, and Colorado locations," *Solar Energy*, vol. 110, pp. 736–744, 2014, doi: 10.1016/j.solener.2014.10.017.

[2] I. Jayawardene et al., "Comparative analysis and trends in performance of identical PV panels under different climatic conditions," *Solar Energy*, vol. 307, art. 114349, 2026, doi: 10.1016/j.solener.2026.114349.

[3] J. R. Angulo et al., "Yield and performance analysis of PERC, HIT, and CIGS photovoltaic systems in five Peruvian city-climates," *Solar Energy*, vol. 304, art. 114204, 2026, doi: 10.1016/j.solener.2025.114204.

[4] M. Kivambe et al., "Comprehensive assessment of performance and reliability of PERC, TOPCon and SHJ modules in desert climates," *Solar Energy*, vol. 295, art. 113555, 2025, doi: 10.1016/j.solener.2025.113555.

[5] G. Friesen et al., *Optimisation of Photovoltaic Systems for Different Climates*, IEA PVPS Task 13, Rep. T13-32:2025, 2025, doi: 10.69766/QSYC8858.

[6] D. Moser et al., *Photovoltaic Project Decisions: Quality, Performance, and Economic Value*, IEA PVPS Task 13, Rep. T13-36:2026, 2026, doi: 10.69766/SDJU3969.

[7] B. Ghosh and S. Mandal, "A new approach for solar photovoltaic parameter extraction using metaheuristic algorithms from manufacturer datasheet," *IEEE Open J. Instrum. Meas.*, vol. 2, pp. 1–12, 2023, doi: 10.1109/OJIM.2023.3318678.

[8] International Electrotechnical Commission, *IEC 61724-1:2021 Photovoltaic System Performance—Part 1: Monitoring*, Geneva, 2021.

[9] International Electrotechnical Commission, *IEC 61853-1:2011 Photovoltaic (PV) Module Performance Testing and Energy Rating—Part 1: Irradiance and Temperature Performance Measurements and Power Rating*, Geneva, 2011.

[10] International Electrotechnical Commission, *IEC 61853-2:2026 Photovoltaic (PV) Module Performance Testing and Energy Rating—Part 2: Spectral Responsivity, Incidence Angle and Nominal Module Operating Temperature Measurements*, ed. 2.0, Geneva, 2026.

[11] International Electrotechnical Commission, *IEC 61853-3:2018 Photovoltaic (PV) Module Performance Testing and Energy Rating—Part 3: Energy Rating of PV Modules*, Geneva, 2018.

[12] International Electrotechnical Commission, *IEC 61853-4:2018 Photovoltaic (PV) Module Performance Testing and Energy Rating—Part 4: Standard Reference Climatic Profiles*, Geneva, 2018.

[13] J. C. Blakesley et al., "Accuracy, cost and sensitivity analysis of PV energy rating," *Solar Energy*, vol. 203, pp. 91–100, 2020, doi: 10.1016/j.solener.2020.03.088.

[14] T. S. Lyubenova, I. Medina, and E. D. Dunlop, "Climatic Specific Energy Rating Analysis of Outdoor PV Field Data," *Progress in Photovoltaics*, vol. 33, no. 12, pp. 1365–1377, 2025, doi: 10.1002/pip.3864.

[15] M. J. Rivera Aguilar and C. Reise, "Extended energy rating of photovoltaic modules by outdoor measurements and uncertainty assessment," *Renewable Energy*, vol. 241, art. 122307, 2025, doi: 10.1016/j.renene.2024.122307.

[16] W. Herrmann et al., "Uncertainty of Climate Specific Energy Rating (CSER) of PV Modules in Accordance With IEC 61853," *Progress in Photovoltaics*, vol. 33, no. 10, pp. 1124–1138, 2025, doi: 10.1002/pip.70007.

[17] X. Huang et al., "Measurement repeatability of crystalline silicon photovoltaic modules in the field," *Progress in Photovoltaics*, vol. 33, no. 8, pp. 854–873, 2025, doi: 10.1002/pip.3927.

[18] G. Malkawi, A. Elsayed, A. Alazzam, A. Omari, S. Badreddine, B. Hussein, M. Alhagyan, and A. Altigani, "A GCC Evidence-Calibrated Nonlinear Decision Framework for Photovoltaic Technology Selection Under Coupled Desert Environmental Stress," *Energies*, vol. 19, no. 16, art. 3908, 2026, doi: 10.3390/en19163908.

[19] P. I. Frazier, "A fully sequential elimination procedure for indifference-zone ranking and selection with tight bounds on probability of correct selection," *Operations Research*, vol. 62, no. 4, pp. 926–942, 2014, doi: 10.1287/opre.2014.1282.

[20] C.-H. Chen, D. He, M. Fu, and L. H. Lee, "Efficient simulation budget allocation for selecting an optimal subset," *INFORMS Journal on Computing*, vol. 20, no. 4, pp. 579–595, 2008, doi: 10.1287/ijoc.1080.0268.

[21] G. Zhang, Y. Peng, J. Zhang, and E. Zhou, "Asymptotically optimal sampling policy for selecting top-m alternatives," *INFORMS Journal on Computing*, vol. 35, no. 6, pp. 1261–1285, 2023, doi: 10.1287/ijoc.2021.0333.

[22] G. Keslin et al., "Ranking and contextual selection," *Operations Research*, vol. 73, no. 5, pp. 2695–2707, 2025, doi: 10.1287/opre.2023.0378.

[23] M. Swaroop, T. Krishnamurti, and B. Wilder, "Distributionally Robust Feature Selection," *Advances in Neural Information Processing Systems*, vol. 38, 2025, doi: 10.52202/085713-2117.

[24] H. Hanada et al., "Distributionally Robust Safe Screening," arXiv:2404.16328, 2024.

[25] C. D. Rodríguez-Gallegos et al., "Real Testing Conditions (RTC) for photovoltaic systems: A climate-responsive framework based on the TruePower™ Alliance," *Solar Energy*, vol. 316, art. 114944, 2026, doi: 10.1016/j.solener.2026.114944.

[26] Sandia National Laboratories PV Performance Modeling Collaborative, "2021 Blind PV Modeling Comparison: IEC-61853-1 Matrix Data for Nine PV Modules," dataset and module test reports, https://pvpmc.sandia.gov/datasets/2021-blind-pv-modeling-comparison/, accessed 2026.

[27] W. Marion et al., "Data for Validating Models for PV Module Performance," U.S. Department of Energy Data Explorer, 2021, doi: 10.21948/1811521.

[28] W. Marion et al., *User's Manual for Data for Validating Models for PV Module Performance*, NREL/TP-5200-61610, 2014, doi: 10.2172/1130632.

[29] IEA PVPS Task 13, "Module Validation Dataset," one-year outdoor series (2018–2019), hosted by Sandia National Laboratories PV Performance Modeling Collaborative (PVPMC), https://pvpmc.sandia.gov/datasets/iea-pvps-task-13-module-validation-dataset/, accessed 2026.

[30] A. Driesse, M. Theristis, and J. Stein, "PV Module Operating Temperature—Data and Resources," dataset, 2023, doi: 10.21948/2204673.

[31] I. Jayawardene and D. Roman, "PV and Weather Datasets from Demosites in Oslo, Touzer, and Sevilla," Zenodo, ver. 2, 2023, doi: 10.5281/zenodo.10423357.

[32] N.-E. Id Omar, L. Boukhattem, F. Oudrhiri Hassani, A. Bennouna, and A. Oukennou, "Data of a PV plant implemented in hot semi-arid climate," *Data in Brief*, vol. 34, art. 106756, 2021, doi: 10.1016/j.dib.2021.106756.

[33] N.-E. Id Omar *et al.*, "Data of three silicon-based PV technologies implemented in hot semi-arid climate," *Mendeley Data*, ver. 1, 2020, doi: 10.17632/b2s4ycc2gz.1.

[34] J. Barry *et al.*, "Dynamic model of photovoltaic module temperature as a function of atmospheric conditions," *Advances in Science and Research*, vol. 17, pp. 165–173, 2020, doi: 10.5194/asr-17-165-2020.

[35] J. Barry *et al.*, "Supplement to Dynamic model of photovoltaic module temperature as a function of atmospheric conditions," *Zenodo*, ver. 1, 2020, doi: 10.5281/zenodo.3958820.

[36] S. Alessandrini, "Real operating data of a photovoltaic system installed at Area Science Park—Trieste—Italy," *Zenodo*, 2022, doi: 10.5281/zenodo.7115550.

[37] C. D. de Jesus, R. N. Ferreira, and E. P. de Aguiar, "BR-PVGen—The Brazilian Photovoltaic Generation Dataset," *Zenodo*, ver. 1, 2026, doi: 10.5281/zenodo.21511487.

[38] A. Ferrero and D. Petri, "Measurement uncertainty and information enabled decision making," *IEEE Open J. Instrum. Meas.*, vol. 4, 2025, doi: 10.1109/OJIM.2025.3626105.

[39] X. Huang, H. Wang, H. Han, T. Guo, H. Niu, Y. Zhu, F. He, and H. Yang, "Measurement reproducibility of STC-corrected maximum power of photovoltaic modules under dynamic outdoor conditions: definition, calculation, and application in power degradation evaluation," *Renewable Energy*, vol. 269, art. 125870, 2026, doi: 10.1016/j.renene.2026.125870.